Computer Networking Project 1

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Build & Deploy a Public Website

🎯 Project Objective

Build a **publicly accessible website** and demonstrate how **networking concepts** (DNS, IP addressing, protocols, and security) are applied in real deployments.

🔧 Project Tasks

1. Domain & Hosting

- Choose a **hosting service** (free allowed):
  - GitHub Pages (recommended)
  - Netlify / Cloudflare Pages / Vercel
- Use:
  - A free subdomain (e.g., username.github.io)
  - OR a temporary/custom domain

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2. Website Setup

- Required:
  - HTML + CSS
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  - JavaScript
  - Frameworks (React, Vue, etc.)

Minimum content

- Home page (project title + team)

- Networking Concepts page
- Optional About/Contact page

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3. Networking Requirements (Required Evidence)

You must demonstrate:

- **DNS**

  Domain resolution (nslookup or dig)

- **IP Addressing**

  IPv4 / IPv6 resolution

- **Protocols**

  HTTP vs HTTPS, headers, status codes

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4. Documentation

Submit a **PDF report** including:

- Website link
- GitHub repo link
- Setup steps
- Networking concepts used
- Challenges + solutions
- Screenshots / command outputs

Implement at least one:

- HTTPS enforcement
- Firewall / WAF rule
- Access control
- Security headers

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5. Submission

- One PDF report per group
- GitHub repository link
- Website link
- Group member names + roles

Grading Checklist (Fast & Practical)

Website (25 pts)

☐ Loads publicly
☐ Pages render correctly
☐ No broken links

Hosting & Domain (20 pts)

☐ Hosting configured
☐ Domain/subdomain resolves correctly

Networking Evidence (35 pts)

☐ DNS output shown (15 pts)
☐ IP address explanation (10 pts)
☐ HTTP/HTTPS headers or devtools (10 pts)

Documentation (10 pts)

☐ Clear steps
☐ Screenshots / outputs included
☐ Team roles explained

Implementation (10 pts)

☐ HTTPS / security feature implemented
☐ Evidence provided

Instructor Notes

Difficulty Level

- *Easy: GitHub Pages + default HTTPS*
- *Medium: Custom domain + DNS records*
- *Hard: CDN + firewall rules + access control*

Common Pitfalls to Watch For

- *Confusing OSI vs TCP/IP*
- *Listing DNS without evidence*
- *Screenshots without explanation*
- *"HTTPS enabled" with no proof*

One Sentence Students Should Say Correctly:

"Our website is hosted publicly, DNS resolves the domain to an IP address, and communication occurs over HTTPS using TCP/IP."